WEEK 1 REPORT

TEAM FORMATION & DATA HUNTING

MONEYBALL – IPL PLAYER VALUATION

DATA SCIENCE AND VISUALIZATION | 24ADI204 | DATA-VIZ HACK-A-BIT (2026–2027)

Team Number: 05

Team Member	Register Number
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Prepared from the executed Week 1 Google Colab notebook and the recorded outputs.

1. Introduction

Moneyball – IPL Player Valuation explores Indian Premier League (IPL) player performance information together with auction-related information. The project aims to examine how player statistics and available valuation fields can be studied using data science techniques and communicated through a dashboard.

Week 1 establishes the project foundation. The executed notebook imports Python libraries, loads the merged IPL dataset, previews records, checks dimensions and column names, inspects data types and summary statistics, and demonstrates introductory Pandas and NumPy operations.

2. Problem Statement

IPL player performance and auction-related information are represented by several statistical and descriptive fields. Examining these fields together can help explore player performance and available valuation information. The project begins by organizing the available data and understanding its structure before deeper analysis in later weeks.

3. Project Objectives

• Load the prepared IPL master dataset into Google Colab.

• Understand the dataset's rows, columns, field names and data types.

• Use Pandas to read and inspect tabular data.

• Use NumPy to demonstrate basic numerical operations.

• Establish a documented baseline for subsequent data science and dashboard work.

4. Team Formation and Responsibilities

The team details below are taken from the supplied team-members slide. Individual task allocations were not specified in that slide; therefore, no individual roles are assigned in this report.

Team Member	Register Number
Divyesh D S	25BAD022
Kavi Prakash M	25BAD043
Mithuin P S	25BAD057
Madhan P M	25BAD503
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Team Number: 05 | Team Size: 5 members.

5. Dataset Selection and Description

The notebook loads **IPL_Moneyball_Merged_2008_2026.csv**. The executed output shows 3,311 records and 27 columns. The dataset contains season and player identifiers, batting and bowling performance measures, player/team descriptors, acquisition details and price-related fields.

Measure	Observed result
Rows	3,311
Columns	27
Distinct values in player column	794
Distinct seasons	19

Season range	2008–2026

5.1 Column names

The 27 columns reported in the notebook are: season, player, runs, balls_faced, innings, fours, sixes, dismissals, balls_bowled, runs_conceded, wickets, matches_bowled, auction_name, team, role, nationality, status, acquisition_type, base_price_lakh, final_price_lakh, bat_style, bowl_style, player_full_name, strike_rate, batting_avg, economy and bowling_avg.

5.2 Original dataset sources and merging

The uploaded notebook reads the already-merged master CSV. It does not record the three original dataset filenames, source URLs, merge keys or merging code. These details must be inserted from the team's actual data-preparation records; they are not assumed or invented here.

Original dataset filenames: [Add the three exact filenames]. Source links: [Add actual URLs]. Merge procedure and join keys: [Describe the actual procedure used after checking the source files/code].

6. Tools and Libraries

Pandas: Used to load the CSV into a DataFrame and inspect tabular data using head(), tail(), sample(), shape, columns, info(), describe() and dtypes.

NumPy: Used for numerical array operations. The notebook creates a sample array and calculates its sum, mean, maximum and minimum.

Matplotlib and Seaborn: Imported for future plotting and statistical visualization. The submitted Week 1 notebook does not show a plot generated using these libraries.

7. Implementation and Results

7.1 Library import and dataset loading

The notebook imports pandas as pd, numpy as np, matplotlib.pyplot as plt and seaborn as sns. The output confirms that all libraries were imported successfully. The CSV is loaded with pd.read_csv('IPL_Moneyball_Merged_2008_2026.csv'), and the notebook prints “Dataset loaded successfully!”.

7.2 Dataset preview

head() displays the first five records, tail() displays the last five records, and sample(5) displays five randomly selected records. The first displayed rows include 2008 player records such as A Chopra, A Kumble, A Mishra, A Mukund and A Nehra.

7.3 Dimensions and structure

The dataset shape is (3311, 27), representing 3,311 rows and 27 columns. The notebook also prints the complete list of column names.

7.4 DataFrame information and data types

The info() output reports 3,311 entries, with index values 0 to 3310. It reports 11 integer columns, 6 floating-point columns and 10 object columns. Memory usage is approximately 698.5 KB. The dtypes operation displays the data type of each column.

The info() output shows that some columns have fewer than 3,311 non-null values. For example, role has 893 non-null values, nationality 1,015, base_price_lakh 202 and final_price_lakh 1,831. These are initial observations only; missing-value treatment is not performed in this Week 1 notebook.

7.5 Descriptive statistics

describe() generates summary statistics for numeric columns. For example, runs has a mean of approximately 111.74, a median of 34 and a maximum of 973. final_price_lakh has 1,831 numeric observations in the summary, a mean of approximately 366.02 and a maximum of 2,520.

7.6 NumPy demonstration

The notebook creates the sample array [450, 520, 380, 600, 700]. Its executed results are: sum = 2,650; mean = 530; maximum = 700; minimum = 380. This is an illustrative array and is not calculated from the IPL dataset.

7.7 Distinct player and season counts

The notebook uses df['player'].nunique() and df['season'].nunique(). The outputs are 794 distinct player values and 19 distinct season values. The dataset shape is again displayed as (3311, 27).

8. Initial Observations

• The master dataset contains 3,311 rows and 27 columns, exceeding the stated minimum requirement of 1,000 rows.

• The dataset covers 19 seasons, from 2008 to 2026 as reflected in the notebook output.

• It combines performance fields such as runs, wickets, strike_rate and economy with descriptive and auction-related fields such as team, role, acquisition_type and final_price_lakh.

• Some columns contain missing values; these should be examined in the relevant later data-inspection and cleaning work.

• The NumPy array is an illustrative example separate from the IPL master dataset.

9. Before-and-After Record

Week 1 establishes the initial loaded-data baseline. The uploaded notebook does not show cleaning, missing-value imputation, duplicate removal, outlier treatment, scaling, encoding, feature selection or dashboard transformation. Therefore, no processed before-and-after dataset comparison is claimed for this week. The baseline is the loaded DataFrame with shape (3311, 27).

10. Conclusion

Week 1 established the working environment and loaded the prepared IPL master dataset into Google Colab. Pandas was used to preview and inspect the dataset, while NumPy was introduced through basic numerical calculations. The notebook verified 3,311 rows, 27 columns, 794 distinct player values and 19 distinct seasons. These results provide the starting point for the next stages of the project.

11. Submission Checklist

■ Upload the executed .ipynb notebook to the existing Week_01/Code folder in the team GitHub repository.

■ Upload this PDF to the existing Week_01/Report folder.

■ Add the three original dataset filenames, their actual source links and the verified merge procedure.

■ Keep the same master dataset for the subsequent project weeks.

Report note: Team number and member names/register numbers are taken from the supplied team-members slide. Dataset source details and the exact merge procedure were not present in the uploaded notebook and remain marked for completion.